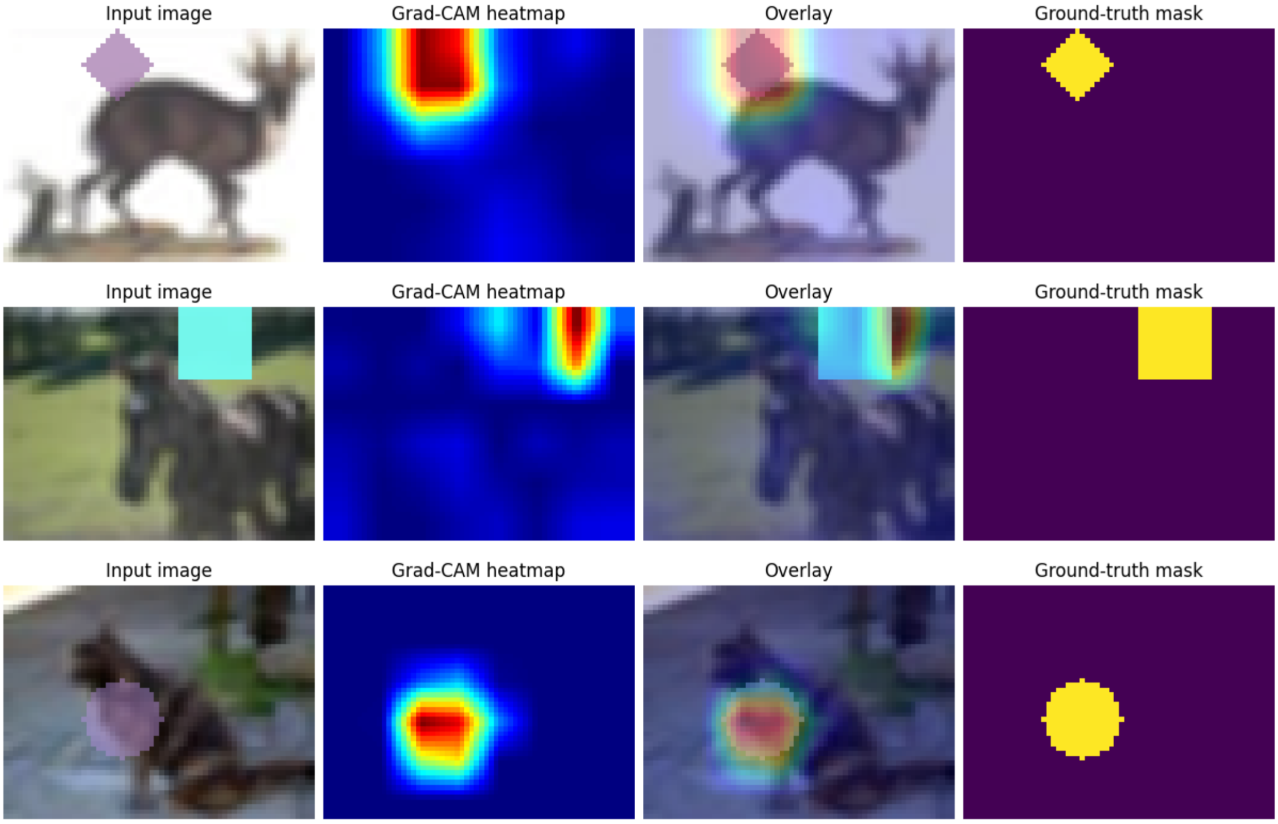
Grad-CAM

Here is examplary visualization of the output of my implementation of Grad-CAM



SAM

Selecting foreground points

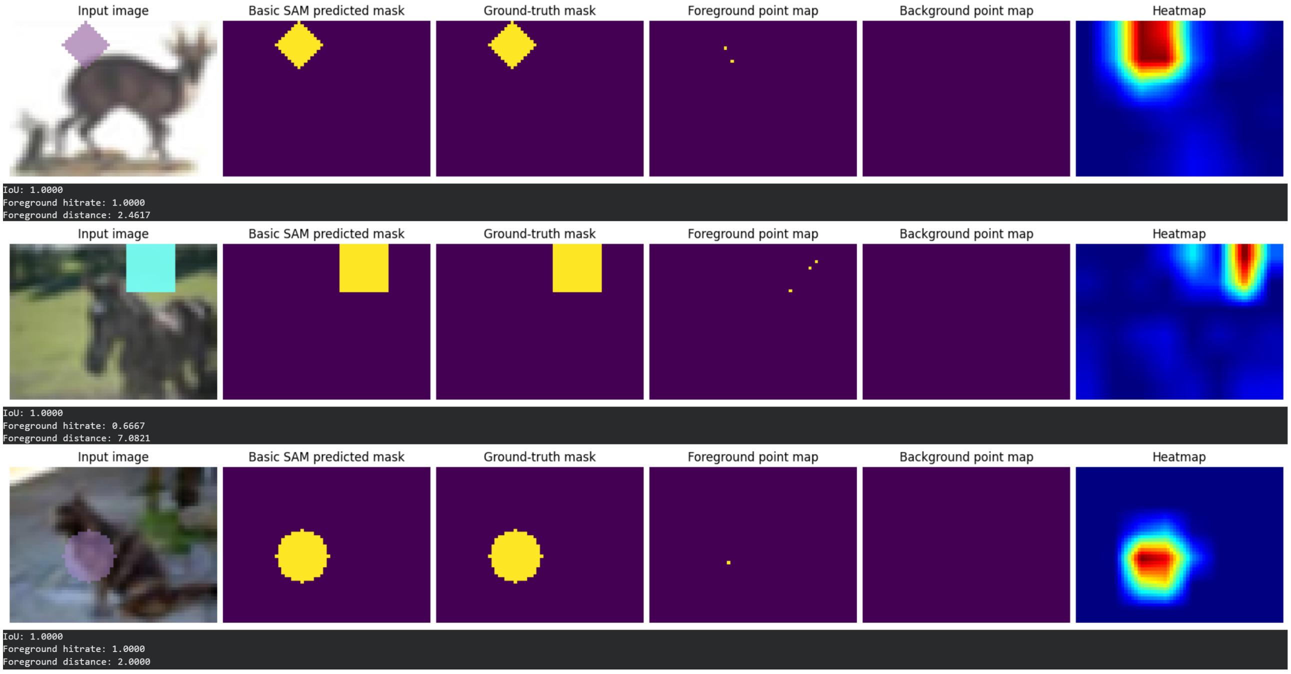
Foreground points are selected from the Grad-CAM heatmap by computing its center of mass. The first point uses the raw Grad-CAM values as weights. The second point is computed in the same way, but with the heatmap values squared. In general, the i-th point is obtained by raising the heatmap values to the i-th power before computing the center of mass. Some points may end up being the same, which is not a problem for the pipeline.

Selecting background points

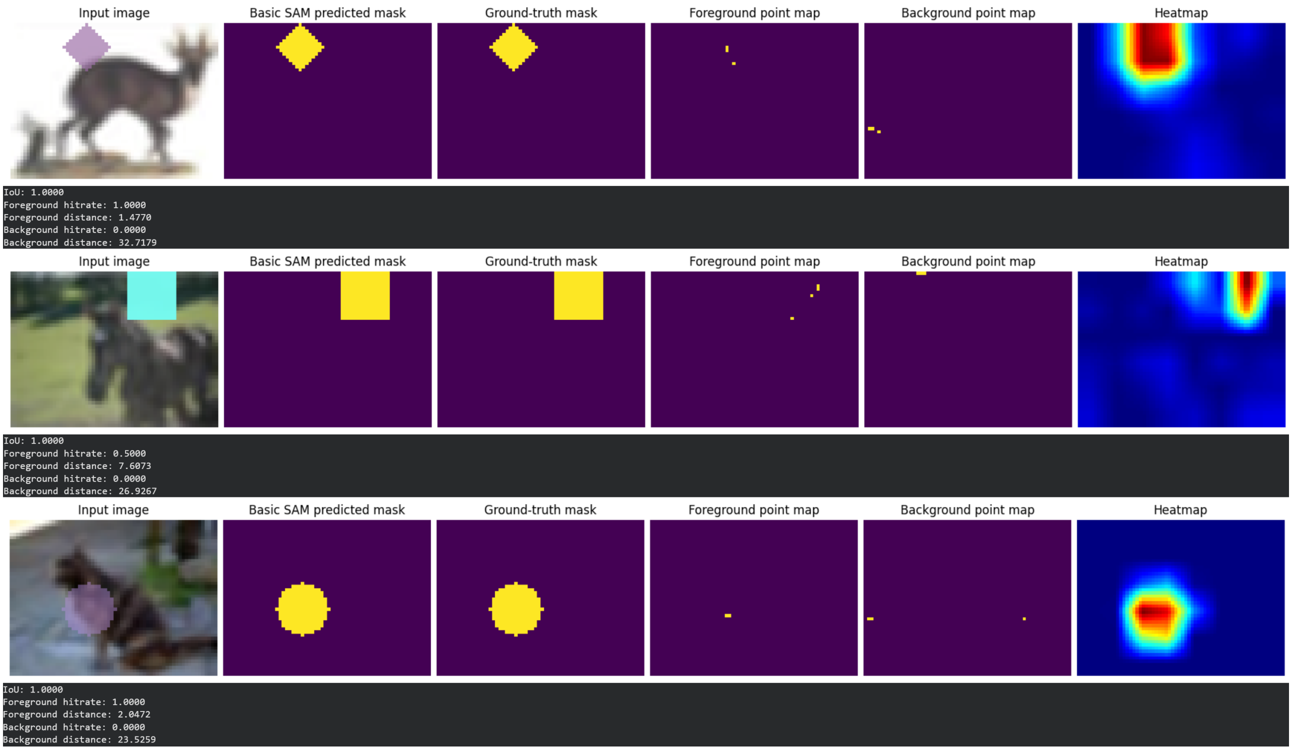
Background points are selected from the Grad-CAM heatmap by simply choosing locations with the lowest values.

Examples

Here is an example output of the pipeline using at most 3 foreground points.



And here using at most 3 foreground points and 3 background points.



Metrics

Table 1: Comparison of metrics

Metric	Only FG points	FG + BG points
Average IOU	0.8526	0.8333
Average foreground hitrate	0.8027	0.8073
Average foreground distance	4.8564	4.7388
Average background hitrate	-	0.0073
Average background distance	-	32.1838

Potencial areas of improvement

To further improve model's performance I would change the strategy of picking background points, because surprisingly current approach lowers IoU. Additionally, experimenting with layers that are considered in Grad-CAM could help. Finally, improving the shape classification model itself might lead to more accurate and stable saliency maps.

An interesting extension of this work would be to consider a more challenging task where multiple shapes can appear and overlap within a single image.